Feasibility Study

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Feasibility Study

- A feasibility study is a study made before committing to a project
- A feasibility study leads to a decision:
  - go ahead or do not go ahead
  - think again
- In production projects, the feasibility study often leads to a resource request
- A feasibility study often ends up often in the form of a proposal or report

Why are Feasibility Studies Difficult?

- **Difficulties**
  - **Benefits are usually very hard to quantify**
  - **Approach is usually ill-defined**
    - Estimates of resources needed and timetable are very rough
  - **Organizational changes may be needed**

- **Therefore, feasibility studies rely heavily on the judgment of experienced people who are often over-enthusiastic**

The Decision Maker's Viewpoint

- A senior member of an organization must decide whether to begin a major software project
  - What information is needed?

- Information needed for the decision maker
  - Client (Who is this project for?)
  - Scope
  - Benefits
  - Technical feasibility
  - Planning and resources
  - Risks and alternatives

Scope

- Scope expresses the boundaries of the system:
  - It will include <list of included functions>
  - It will exclude <list of excluded functions>
  - It depends on <list of dependencies>
  - It replaces <list of functions to be replaced>

- Confusion over scope is a common reason for clients to be dissatisfied with a system
  - "Is that all you planned to do?"
  - "But I assumed that you were going to do XYZ."
  - "I can't use the system without ABC."

Benefits

- Why is this project proposed?
  - Can you quantify the benefits?
  - How much will it contribute to the corporate's revenue?

- Examples
  - Create a marketable product
  - Improve the efficiency of an organization (e.g., save staff)
  - Control a system that is too complex to control manually
  - New or improved service (e.g., faster response to customers)
  - Safety or security

Technical Feasibility

- A feasibility study needs to demonstrate that the proposed system is technically feasible, and this requires
  - a rough outline of the requirements
  - a possible system design (e.g., database, distributed, etc.)
  - estimates of numbers of users, data, transactions, etc.
  - possible choices of software to be acquired or developed

- These very rough numbers are fed into the provisional plan that is used to estimate the staffing, timetable, equipment needs, etc
  - The technical approach actually followed may be very different

Planning and Resources

- The feasibility study should include an outline plan:
  - Estimate the staffing and equipment needs, and the preliminary timetable
  - Identify major decision points
  - Identify interactions with and dependences on external systems
  - Provide a preliminary list of deliverables and delivery dates

Risks and Alternatives

- A feasibility study should identify risks and alternatives

- Risks
  - What can go wrong?
  - How will problems be identified (visibility)?
  - Are there fall-back options?

- Alternatives
  - Continue with current system, enhance it, or create new one?
  - Develop in-house, or contract out? (How will a contract be managed?)

Feasibility Report

- A written document
  - For a general audience: client, financial management, technical management, etc.
  - Short enough that everybody reads it
  - Long enough that no important topics are skipped
  - Details are often included in supporting documents

- It should be a well written, well presented document

Feasibility Report

- **The purpose of a feasible study is to establish if a project is feasible, at reasonable cost, within the planned period**
- **The feasibility report should conclude with recommendations about whether to proceed, but the final decision is made jointly by the client and the development team**
- **In previous years, many reports have had the following problems:**
  - **The report is vague about the scope**
    - Without a clear definition of scope, it is not clear that the project is feasible
  - **The plan does not describe the activities in enough detail to estimate the effort convincingly**
  - **The projects is too ambitious**
    - The report needs to describe how will you monitor the progress and adjust the scope if necessary


Q & A

Feasibility Study?

- 예시
  - 게임회사에서 게임 아이템 거래를 활성화하기 위해 블록체인 기반의 암호화폐 토큰**(token)**을 도입하고자 한다면 **feasibility study**는 어떻게 되는가**?**
  - 카카오와 같은 메신저 회사에서 **zoom**과 같은 화상회의 솔루션을 개발하고자 한다면 **feasibility study**는 어떻게 되는가**?**

- 고려 요소
  - **(User)** 사용자는 누구인가**?**
  - **(Scope)** 개발의 범위는 어디까지 봐야 하는가**?**
  - **(Benefits)** 회사에 어떤 이득을 가져다 줄 수 있는가**?**
  - **(Technical feasibility)** 회사의 역량으로 개발이 가능한가**?**
  - **(Planning and resources)** 투입될 개발인원, 비용, 기간은 어떠한가**?**
  - **(Risks and alternatives)** 어떤 리스크가 있을 수 있는가**?**